

Association of Sugar Restriction in Utero Through Age 2 Years on Dementia Risk Later in Life

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Abstract

Background and Objectives

Excess sugar intake in early life may affect long-term brain health, but evidence for dementia is limited. We used the abrupt end of UK sugar rationing in September 1953 as a natural experiment to test whether exposure to sugar rationing during different windows within the first 1,000 days from conception was associated with adult risk of all-cause dementia, Alzheimer disease (AD), and vascular dementia (VaD).

Methods

We analyzed UK Biobank participants born around the end of UK sugar rationing. Exposure was classified as rationing in utero only, in utero plus the first year of life, in utero plus 1–2 years of life, or no exposure. Incident dementia was identified from linked ICD-10 records. MRI of the brain and cognitive function were assessed in the imaging subcohort. Adjusted Cox and Gompertz models estimated HRs and 95% CIs, Fine–Gray models accounted for competing risk, and mediation analyses evaluated type 2 diabetes and hypertension.

Results

Among 64,737 participants included in the analysis, the mean age at recruitment was 54.6 years, and 56.4% were women; 40,963 were exposed to sugar rationing during fetal and/or early-childhood life, and 23,774 were unexposed. Compared with unexposed individuals, sugar rationing in utero plus the first year of life was associated with lower hazards of all-cause dementia (HR, 0.79; 95% CI 0.66–0.94) and AD (HR, 0.77; 95% CI 0.59–1.00). Similar or slightly stronger associations were observed for exposure in utero plus 1–2 years (all-cause dementia: HR, 0.77; 95% CI 0.63–0.95; AD: HR, 0.72; 95% CI 0.53–0.98). Exposure in utero plus 1–2 years was associated with delayed onset of all-cause dementia by 2.55 years, AD by 2.87 years, and VaD by 2.49 years. Early-life sugar rationing was also associated with higher total gray matter volume (β , 3.27; 95% CI 0.46–6.07), lower white matter hyperintensity volume (β , –0.64; 95% CI –0.97 to –0.31), and better performance in processing speed and reasoning. Incident type 2 diabetes and hypertension jointly mediated 25.5% of the association.

Discussion

Sugar restriction in the first 1,000 days was associated with lower dementia hazards, delayed onset, and more favorable brain-health profiles. These results support early-life sugar reduction as a potential strategy for dementia prevention.

Introduction

The “fetal origins of disease” hypothesis in the cognitive domain is preliminarily supported by animal studies showing that dams fed a high-sugar diet during gestation and lactation exhibit

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Supplementary Material

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Glossary

AD = Alzheimer disease; **APOE** = apolipoprotein E; **BMI** = body mass index; **CPI** = consumer price index; **CVD** = cardiovascular disease; **DAG** = directed acyclic graph; **HR** = hazard ratio; **ICD-10** = International Classification of Diseases, 10th Revision; **MET** = metabolic equivalent task; **NFS** = National Food Survey; **NHS** = National Health Service; **ONS** = Office for National Statistics; **SHR** = subdistribution hazard ratio; **VaD** = vascular dementia.

impaired long-term memory performance in adult female offspring rats (more details are shown in the eAppendix 1).¹ Human evidence is also accumulating. Maternal sucrose consumption has been inversely associated with mid-childhood nonverbal scores (−1.5 points per 15 g/d).² Infant cognitive development scores at 24 months were inversely correlated with maternal fructose consumption during the first postnatal month (beta = −0.08).³ Berger et al.⁴ suggested that maternal total and added sugar intake during the second trimester is significantly associated with brain tissue organization in infants. In addition, children exposed to maternal diabetes show lower whole-brain cortical thickness compared with unexposed children (mean: exposed = 2.725 mm vs unexposed = 2.732 mm).⁵ However, despite the above evidence, the association between early-life sugar restriction and dementia risk, neuroimaging outcomes, and cognitive function in adulthood remained unclear.

Our study leveraged the UK's sugar rationing policy as a natural experiment, which was introduced in July 1942 as part of a broader food rationing effort during World War II (more details are shown in the eAppendix 2 in the supplementary material).⁶ Previous famine and undernutrition cohorts documented the harmful effects of severe deprivation within the Developmental Origins of Health and Disease framework.^{7–9} By contrast, Britain's sugar rationing limited added sugars without broad caloric deprivation, providing a unique contrast to these extreme exposures while still offering insights into the role of early-life nutrition in shaping long-term health outcomes.

This is an observational study using data from the UK Biobank, a large prospective longitudinal cohort in the United Kingdom. Participants were recruited in 2006–2010 and have been followed through linked health records for incident outcomes. In addition to baseline questionnaires, interviews, and physical measurements, a subset later attended an imaging visit with standardized brain MRI and cognitive testing. This study aimed to evaluate the long-term effects of restricted sugar consumption during the first 1,000 days of life on cognitive health in adulthood. Specifically, we assessed the risks of all-cause dementia, Alzheimer disease, and vascular dementia by comparing individuals who experienced limited sugar intake in utero and infancy due to the rationing system with those who had higher sugar consumption after the policy ended. Neuroimaging and cognitive function data were included to explore potential subclinical cognitive changes. We further conducted mediation analyses to evaluate whether incident type 2 diabetes, hypertension, and birth weight may

lie on the pathway between early-life sugar restriction and dementia and to quantify the proportion of the association mediated by these factors.

Methods

Study Design and Participants

The cessation of sugar rationing in September 1953 can be considered a natural experiment because it marked a sudden, large-scale policy change affecting the population, without prior planning or manipulation by researchers. Individuals born before and after this date were subject to different levels of sugar exposure during critical periods of development. Those born before June 1954 experienced sugar rationing (including in-utero exposure), while those born after had access to unrestricted sugar. This creates a natural distinction between groups, based on birth date. We use an event-study approach to investigate how the restriction of sugar intake during the first 1,000 days after conception affects dementia and cognitive function in later years. This quasiexperimental design has been used by Gracner et al.¹⁰ in a previous study.

The National Food Survey (NFS) is a UK government-led, nationally representative household survey of food acquisition and consumption,^{10–12} in which participating households (typically the main food purchaser/preparer) recorded all foods brought into the home over a 7-day diary period, supported by trained interviewers and standardized coding procedures. Following the approach of Gracner et al.,¹⁰ we used the historic NFS/Family Food reports to derive national quarterly dietary trends (e.g., sugar and fat intake) and to construct the “social average fat intake” covariate for participants' birth period. Annual Food Price Index and related socioeconomic indicators were extracted from the official historic NFS/Family Food statistical reports (see eMethods).¹³ Annual consumer price index (CPI) was obtained from the Office for National Statistics (ONS) historical CPI estimates for 1950–1988 (see eMethods).¹⁴

UK Biobank is a large population-based cohort of ~500,000 adults aged 40–70 years recruited across the United Kingdom in 2006–2010. Participants completed questionnaires and interviews, underwent physical measurements, and provided biological samples; subsets later completed brain MRI and cognitive testing. For this analysis, sugar-rationing policy exposure was assigned using date of birth relative to the rationing period, and outcomes/covariates were obtained from baseline assessments and linked health records (and brain

MRI/cognitive assessments for neuroimaging/cognition analyses).¹⁵

UK Biobank enrolls community-dwelling adults and is not a birth cohort; hence, our analyses condition on surviving and attending the baseline assessment. Our primary scientific question was: is early-life exposure to sugar rationing associated with incident dementia and cognitive function in mid-to late adulthood? Therefore, we excluded participants with prevalent dementia at baseline, defined as any dementia diagnosis recorded before the baseline assessment date in linked health records, to ensure analyses of incident dementia during follow-up. In addition, participants from multiple births (e.g., twins or triplets), identified from baseline report, were excluded to reduce heterogeneity related to systematically different early-life conditions (e.g., prematurity and lower birth weight) and to avoid potential nonindependence where siblings might both be enrolled.

Definition of Early-Life Exposure to Sugar Rationing

To define the 1,000-day period around the cessation of sugar rationing, we approximate each quarter as lasting about 91 days. With 11 quarters spanning from the beginning of 1951 to mid-1954 (approximately 1,001 days), we establish the 1,000-day period from October 1, 1951, to June 30, 1954. Adults born after July 1, 1954, are considered to have never experienced sugar rationing, meaning they were not exposed to the rationing policy. Our method is based on the principles of regression discontinuity design.¹⁶ By focusing on those who were pregnant around the end of the rationing period, we assume that other early-life health conditions were similar across both groups and remained largely unchanged. This narrow timeframe surrounding the end of rationing allows us to hypothesize that, had there been no rationing, the health outcomes and disease rates for individuals conceived before or after September 1953 would be relatively stable. For those born after the rationing ended, cognitive outcomes would remain unaffected by their conception date.

Rationing exposure was classified into several groups based on its duration: no exposure, exposure only in utero, and exposure in utero and up to 6, 12, 18, or 24 months after birth. We created a comparison group made up of individuals born after the conclusion of rationing and analyzed outcome differences across 6-month birth cohorts. This strategy helped mitigate any potential biases related to broader time trends or improvements in disease detection that might lead to earlier diagnosis in adults not exposed to rationing.¹⁷ If such biases existed, the risk of chronic diseases would increase over time. Our main assumption was that, once control factors were accounted for, the disease risk for adults born after December 1954, who were not exposed to rationing, would be consistent, with their hazard ratios (HRs) staying close to 1.

Broader categories were also formed, including exposure solely in utero, exposure in utero and the first year after birth,

exposure in utero plus 1 to 2 years after birth, or no exposure. The distribution of rationing exposure and birth dates throughout the study period are depicted in Figure 1.

Covariates

At the baseline assessment (2006–2010), through electronic interactive participant questionnaires, the UK Biobank collected information on age, sex, place of birth (England, Wales, Scotland), education level (Below A-levels, A-levels (high school), College/university, Professional/other), smoking status (Never, Former, Current), alcohol consumption frequency (<3 times/week, ≥3 times/week), physical activity, household income (<18,000, 18,000–30,999, 31,000–51,999, 52,000–100,000, >100,000£), and the Townsend deprivation index. Data on medical conditions, including cardiovascular disease, digestive diseases, kidney disease, and liver disease, were obtained from health records. Body mass index (BMI) was measured during physical assessments. Information on family history and maternal conditions, such as smoking during pregnancy, breastfeeding, and birth weight, was collected through interviews conducted by professionally trained interviewers.

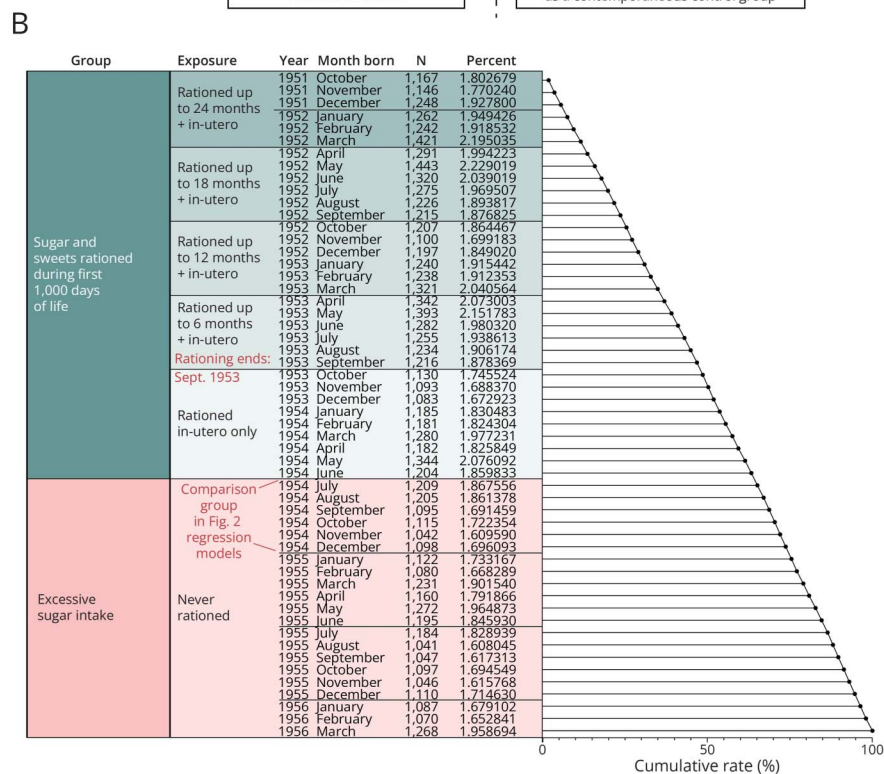
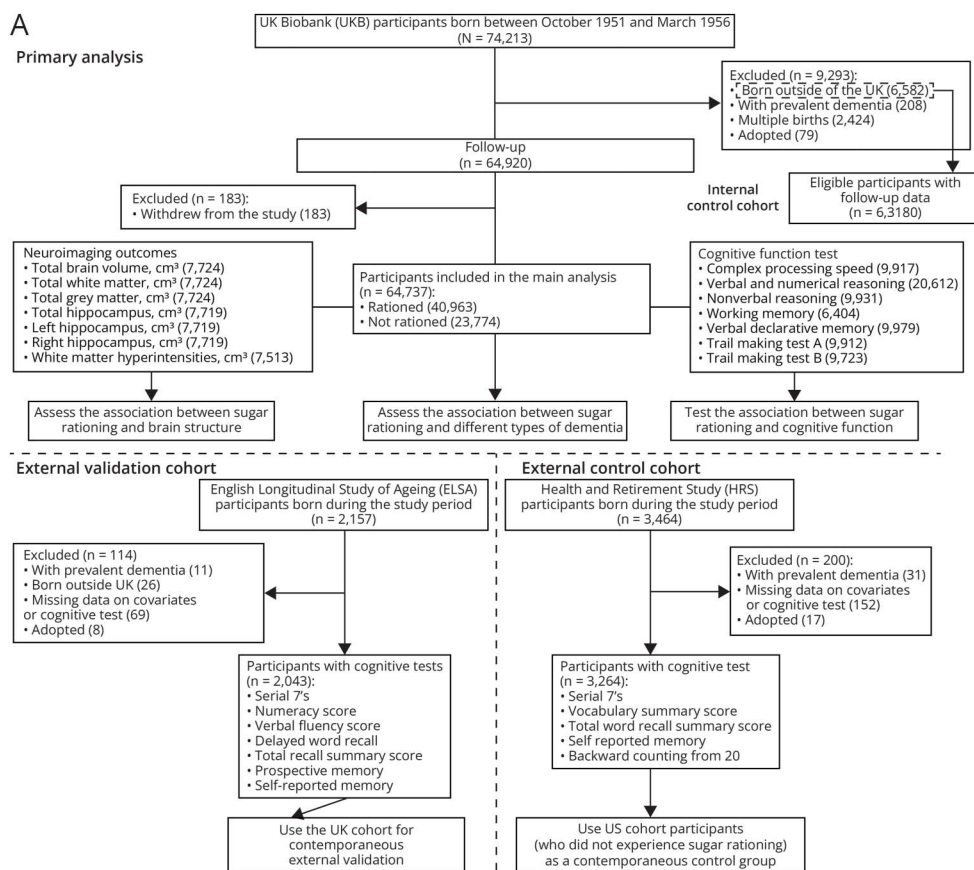
Physical activity was self-reported at the baseline assessment using touchscreen lifestyle questionnaire and was operationalized as the UK Biobank-derived summed metabolic equivalent task (MET) minutes per week for all activity. BMI was calculated using participants' weight and height, applying the formula: $BMI = \text{weight (kg)} / \text{height (m}^2\text{)}$. The Townsend deprivation index is a measure of social deprivation based on factors such as unemployment, car ownership, and housing tenure in the participant's area.¹⁸ Hypertension was defined as having a blood pressure reading ≥140/90 mm Hg or being on antihypertensive treatment. Diabetes was classified according to a fasting blood glucose level ≥7.0 mmol/L or glycated hemoglobin (HbA1c) ≥6.5% or current use of insulin or oral hypoglycemic agents. High cholesterol was identified if total cholesterol levels were ≥6.2 mmol/L or participants were on lipid-lowering medications. The analysis method for the apolipoprotein E ε4 (APOE ε4) genotype is described in the eMethods in the supplementary material.

Assessment of Dementia and Placebo Outcomes

In this study, the primary outcomes include all-cause dementia, Alzheimer disease, and vascular dementia. The dates and causes of death were obtained through linkage to the National Health Service (NHS) death registers for England and Wales, as well as the NHS Central Register for Scotland. In addition, hospitalization dates and causes were determined by linking data from the Scottish participants' incidence records and hospital morbidity statistics from England and Wales participants. Dementia was defined using International Classification of Diseases, 10th Revision (ICD-10) codes, with detailed information presented in eTable 1.

For our placebo analysis, we examined diseases that are less likely to be influenced by in-utero or early-life sugar exposure,

Figure 1 (A) Flowchart Depicting the Study Cohort. (B) Distribution of Birth Samples in the UK Biobank by Calendar Month and Their Exposure to Sugar Rationing



The sugar-rationed group is shown in green, while the group unexposed to sugar rationing is shown in pink. The first group of individuals who were never exposed to sugar rationing has been labeled and used as the control group in Figure 2.

including type 1 diabetes (defined by ICD-10 code E10) and melanoma (defined by ICD-10 code C43). The follow-up period for all participants started at recruitment and ended when the outcome of interest was diagnosed, when the participant died, was lost to follow-up, or when the study concluded on July 1, 2023, whichever came first.

Assessment of Neuroimaging Outcomes and Cognitive Function

In the UK Biobank imaging study (initiated in 2014 with the aim of re-inviting ~100,000 participants), cognitive assessments were conducted at the imaging visit using tests that demonstrated reliability and validity, covering a range of cognitive domains. Brain structure assessments were conducted using T1-weighted and T2-weighted fluid-attenuated inversion recovery (T2-FLAIR) MRI scans.¹⁹ T1 scans enabled the measurement of brain tissue and structural volumes, while T2-FLAIR scans facilitated the identification of pathologic changes such as white matter hyperintensities.²⁰ Selected imaging markers to indicate brain health included total brain volume, total white matter volume, total gray matter volume, volume of bilateral hippocampus, and volume of white matter hyperintensities. We used UK Biobank imaging-derived phenotypes, which are provided as precomputed numerical variables in the UK Biobank database and generated by the UK Biobank brain imaging processing pipeline (including FAST/SIENAX-style volumetry, FIRST subcortical segmentation, and BIANCA-derived white matter hyperintensities measures, et al).

Cognitive assessments were conducted using tests that demonstrated reliability and validity,²¹ covering a range of cognitive domains. These included executive function, assessed using the Trail Making Test Parts A and B; verbal and numerical reasoning (fluid intelligence); working memory, measured through the backward digit span task; complex processing speed, evaluated with the symbol digit substitution test; verbal declarative memory, tested with paired associate learning; and nonverbal reasoning, measured using the matrix pattern completion test. These tests were carefully chosen to provide comprehensive insights into different aspects of cognitive function. For detailed evaluation methods, please refer to eTable 2.

External Validation and Control Cohorts

We used 3 cohorts for comparison: an internal contemporaneous control group from the UK Biobank, an external validation group from the English Longitudinal Study of Ageing,²² and an external control group from the Health and Retirement Study.²³ Details are shown in the eMethods in the supplementary material.

Statistical Analysis

Continuous variables are expressed as means $\pm$ SD, and categorical variables are presented as frequencies (percentages). Missing data is outlined in eTable 3. To address potential bias, we performed multiple imputations using chained equations

across 20 data sets, with the imputation process described in the eMethods in the supplementary material. The χ^2 test was used to calculate *p* values for categorical variables, and the Mann-Whitney *U* test was applied to continuous variables to compare differences between rationed and nonrationed groups. A comprehensive overview of the statistical analysis methods can be found in the eMethods in the supplementary material. Cox proportional hazards models and Gompertz distribution-based risk models were used to estimate HRs and 95% CI, evaluating the link between sugar rationing and the risk of dementia. The proportional hazards assumption was checked using Schoenfeld residual plots, and no violations were detected. Based on the Akaike Information Criterion and Bayesian Information Criterion,²⁴ the Gompertz distribution was selected as the best-fit model (eTable 4).

We used a directed acyclic graph (DAG) generated via the R package “dagitty,”²⁵ to guide covariate selection (eFigure 1). Model 1 adjusted for age and sex. Model 2 adjusted for DAG-selected covariates (determined at birth) to mitigate endogeneity bias: demographic factors (age, sex, race, birth location, birth month), inflation-adjusted food prices, parental dementia and survival status, early-life exposures (maternal smoking, breastfeeding), social average fat intake during birth year (from NFS), and survey year. Model 3 additionally adjusted for socioeconomic status (household income, Townsend deprivation index), baseline lifestyle factors (smoking, alcohol, physical activity), and medical conditions (cardiovascular disease [CVD]/hypertension/diabetes/digestive/kidney/liver diseases), aiming to better isolate sugar rationing's effect on dementia.

The detailed steps of additional statistical methods and robustness checks are shown in the eMethods in the supplementary material.

All statistical analyses were performed using R version 4.0.2 (R Core Team, Vienna, Austria), with a two-tailed significance level set at *p* < 0.05.

Standard Protocol Approvals, Registrations, and Participant Consents

The UK Biobank has received approval from the North West Multicenter Research Ethics Committee. All participants provided signed consent.

Data Availability

The data can be requested from UK Biobank. This study was conducted under the application No. 97089 of the UK Biobank Resource.

Results

Postrationing Dietary Shifts in the UK

The postrationing trend of increased sugar intake in the UK has already been reported by Gracner et al.¹⁰ We have

recollected and visualized these dietary shifts primarily for clarity and ease of interpretation. According to NFS data, sugar intake rose by over 50% postrationing, while fat consumption showed a modest increase, and most other commodities changed minimally (eFigure 2A). After sugar rationing ended, the total calorie increase was relatively small (about 5% overall; see eFigure 2B). The total calorie changes stemmed chiefly from sugar intake, with a lesser contribution from fat (eFigure 2C). eFigure 2D showed that sugar consumption patterns across the 4 social classes (A, B, C, D) remained remarkably consistent.

Characteristics of the Study Population

Figure 1 presents the study design and cohort selection process, including participant exclusions and exposure categorization. From the UK Biobank, 74,213 participants born between 1951 and 1956 were considered, with 9,476 excluded (including 6,582 born outside the UK, 208 with prevalent dementia, 2,424 with multiple births, 79 adopted, and 108 withdrawn from the study). Among the remaining 64,737 participants, 23,774 did not experience sugar rationing.

Table 1 presents baseline characteristics of participants in a study, comparing those who were rationed (40,963) vs those not rationed (23,774). The rationed group was older (mean age 55.4 vs 53.2, $p < 0.001$). Women accounted for 56.3% of the rationed group and 56.5% of the nonrationed group ($p = 0.579$). In the rationed group, more participants were born between March 1 and May 31, and more were born in England ($p = 0.003$). Educational attainment was lower in the rationed group, and a smaller percentage had a household income over £18,000 ($p < 0.001$). The rationed group had fewer current smokers, and personal medical conditions, such as hypertension and diabetes, were more prevalent in the rationed group.

Association Between Early-Life Sugar Restriction and Dementia

Sugar restriction during the in-utero period plus the first year of life was associated with lower risks of all-cause dementia (HR = 0.79, 95% CI 0.66–0.94) and Alzheimer disease (0.77, 0.59–1.00). Point estimates were slightly lower for the in-utero plus 1–2 years exposure window (all-cause dementia: HR = 0.77, 95% CI 0.63–0.95; Alzheimer disease: 0.72, 0.53–0.98), and the p for trend was lower than 0.05 (Table 2). Vascular dementia showed a comparable pattern, but the associations were not statistically significant. Across all dementia types, HRs were consistently below 1 in the exposed groups.

Figure 2 presents penalized cubic spline plots, with individuals born between July and December 1954 as the reference group. For all 3 outcomes, HRs remained below 1 for individuals exposed to sugar rationing from in utero plus 24 months to those exposed in utero only. As the exposure duration shortened, the protective effect gradually weakened, with HRs approaching 1 and stabilizing during the nonrationing period. For the nonrationed group, no significant differences in disease incidence were found compared with

the reference group, with p values of 0.483, 0.520, and 0.589 for all-cause dementia, Alzheimer disease, and vascular dementia, respectively.

eFigure 3 showed HRs for dementia outcomes by various factors, comparing rationed vs nonrationed participants. A significant interaction was found for sex ($p = 0.038$), with a stronger protective effect in females. No significant interactions were observed for ethnicity, place of birth, *APOE* genotype, parents diagnosed with dementia, maternal smoking around birth, or breastfeeding as a baby. Across the deciles of northern and eastern birth coordinates, sugar rationing demonstrated consistent protective effects (eTable 5).

eTable 6 presents that for all-cause dementia, the delay in onset is 2.55 years (95% CI 2.10–2.99) for those exposed to sugar rationing in utero plus 1–2 years, compared with the nonrationed group. Similarly, Alzheimer disease shows a delay of 2.87 years (95% CI 2.06–3.69), and vascular dementia shows a delay of 2.49 years (95% CI 1.65–3.33) for the same exposure duration.

Associations Between Sugar Rationing and Neuroimaging Outcomes and Cognitive Function

In the UK Biobank imaging subcohort, the mean age at the imaging visit was 61.8 (8.3) years. eFigures 4–5 showed the distributions of neuroimaging and cognitive function metrics, with white matter hyperintensities and verbal declarative memory exhibiting skewed distributions. In model 3, for neuroimaging, the rationed group had higher total grey matter volume (coefficient = 3.27, 95% CI 0.46–6.07, $p = 0.022$) and lower white matter hyperintensities (−0.64, −0.97 to −0.31, $p < 0.001$) compared with the nonrationed group (Table 3). For cognitive function, the rationed group showed better performance in complex processing speed (coefficient = 0.27, 95% CI 0.006–0.54, $p = 0.040$), verbal and numerical reasoning (0.10, 0.01–0.19, $p < 0.05$), nonverbal reasoning (0.13, 0.022–0.25, $p = 0.018$), and Trail Making Test A (−4.53, −8.82 to −0.25, $p = 0.038$).

Sensitivity Analysis and Placebo Tests

A Fine and Gray model was used to adjust for competing risks (eFigure 6). There were 3,191 competing risk events recorded for all-cause dementia, 3,457 for Alzheimer's disease, and 3,503 for vascular dementia. The results are similar to our main analysis, with the rationed group showing a lower incidence of all-cause dementia (subdistribution hazard ratio [SHR] = 0.85, 95% CI 0.73–0.98). SHR was 0.95 (95% CI 0.77–1.18) for Alzheimer disease and 0.84 (95% CI 0.64–1.11) for vascular dementia.

Compared with the nonrationed group, the HRs for placebo outcomes including type 1 diabetes and melanoma were close to 1 at various levels of rationing exposure, indicating negative results (eFigure 7).

In eFigure 8, we found that incident type 2 diabetes and incident hypertension partially mediated 18.0% and 11.6% of the

Table 1 Baseline Characteristics of Rationed and Nonrationed Participants

	Total	Rationed	Not rationed	p Value
No. of participants	64,737	40,963	23,774	
Age at entry, mean (SD), y	54.6 (1.6)	55.4 (1.2)	53.2 (1.0)	<0.001
Women	36,496 (56.4)	23,059 (56.3)	13,437 (56.5)	0.579
Place of birth				
England	55,672 (86.0)	35,320 (86.2)	20,352 (85.6)	0.003
Wales	3,151 (4.9)	2019 (4.9)	1,132 (4.8)	
Scotland	5,914 (9.1)	3,624 (8.8)	2,290 (9.6)	
Birth month				
March 1–May 31	16,948 (26.2)	12,017 (29.3)	4,931 (20.7)	<0.001
June 1–August 31	14,630 (22.6)	8,796 (21.5)	5,834 (24.5)	
September 1–November 30	15,716 (24.3)	9,274 (22.6)	6,442 (27.1)	
December 1–February 28	17,443 (26.9)	10,876 (26.6)	6,567 (27.6)	
White, %	62,275 (96.2)	39,471 (96.4)	22,804 (95.9)	0.005
Education, %				
Below A levels	26,867 (41.5)	17,174 (41.9)	9,693 (40.8)	<0.001
A levels (high school)	20,203 (31.2)	12,462 (30.4)	7,741 (32.6)	
College/university	7,949 (12.3)	5,037 (12.3)	2,912 (12.2)	
Professional/other	9,718 (15.0)	6,290 (15.4)	3,428 (14.4)	
Household income, £				
<18,000	15,405 (23.8)	10,138 (24.7)	5,267 (22.2)	<0.001
18,000–30,999	12,056 (18.6)	7,869 (19.2)	4,187 (17.6)	
31,000–51,999	17,050 (26.3)	10,723 (26.2)	6,327 (26.6)	
52,000–100,000	15,659 (24.2)	9,510 (23.2)	6,149 (25.9)	
>100,000	4,567 (7.1)	2,723 (6.6)	1844 (7.8)	
Townsend deprivation index, mean (SD)	−1.4 (3.0)	−1.5 (3.0)	−1.4 (3.0)	<0.001
Smoking status, %				
Never	36,258 (56.0)	22,742 (55.5)	13,516 (56.9)	<0.001
Former	21,208 (32.8)	13,780 (33.6)	7,428 (31.2)	
Current	7,271 (11.2)	4,441 (10.8)	2,830 (11.9)	
Alcohol consumption frequency, times/wk				
<3	34,991 (54.1)	22,051 (53.8)	12,940 (54.4)	0.143
≥3	29,746 (45.9)	18,912 (46.2)	10,834 (45.6)	
BMI, mean (SD), kg/m²	27.5 (5.0)	27.5 (5.0)	27.5 (5.0)	0.207
Summed MET minutes per week for all activity, IQR	1,800 (798–3,513)	1,804 (794–3,524)	1,798 (809–3,492)	0.964
Personal medical condition				
Cardiovascular diseases	2,390 (3.7)	1,625 (4.0)	765 (3.2)	<0.001
Hypertension	35,993 (55.6)	23,111 (56.4)	12,882 (54.2)	<0.001
High cholesterol	11,328 (17.5)	7,416 (18.1)	3,912 (16.5)	<0.001

Continued

Table 1 Baseline Characteristics of Rationed and Nonrationed Participants (*continued*)

	Total	Rationed	Not rationed	<i>p</i> Value
Diabetes	3,372 (5.2)	2,191 (5.3)	1,181 (5.0)	0.037
Digestive disease	549 (0.8)	366 (0.9)	183 (0.8)	0.107
Kidney disease	1754 (2.7)	1,175 (2.9)	579 (2.4)	0.001
Liver disease	2,286 (3.5)	1,483 (3.6)	803 (3.4)	0.112
Parents' condition				
Parents diagnosed with dementia	8,181 (12.6)	5,420 (13.2)	2,761 (11.6)	<0.001
Parents still alive	10,188 (15.7)	5,435 (13.3)	4,753 (20.0)	<0.001
Birth weight, mean (SD), kg	3.3 (0.5)	3.3 (0.5)	3.3 (0.5)	0.631
Maternal smoking around birth	19,851 (30.7)	12,559 (30.7)	7,292 (30.7)	0.977
Breastfed as a baby	37,802 (58.4)	24,012 (58.6)	13,790 (58.0)	0.128

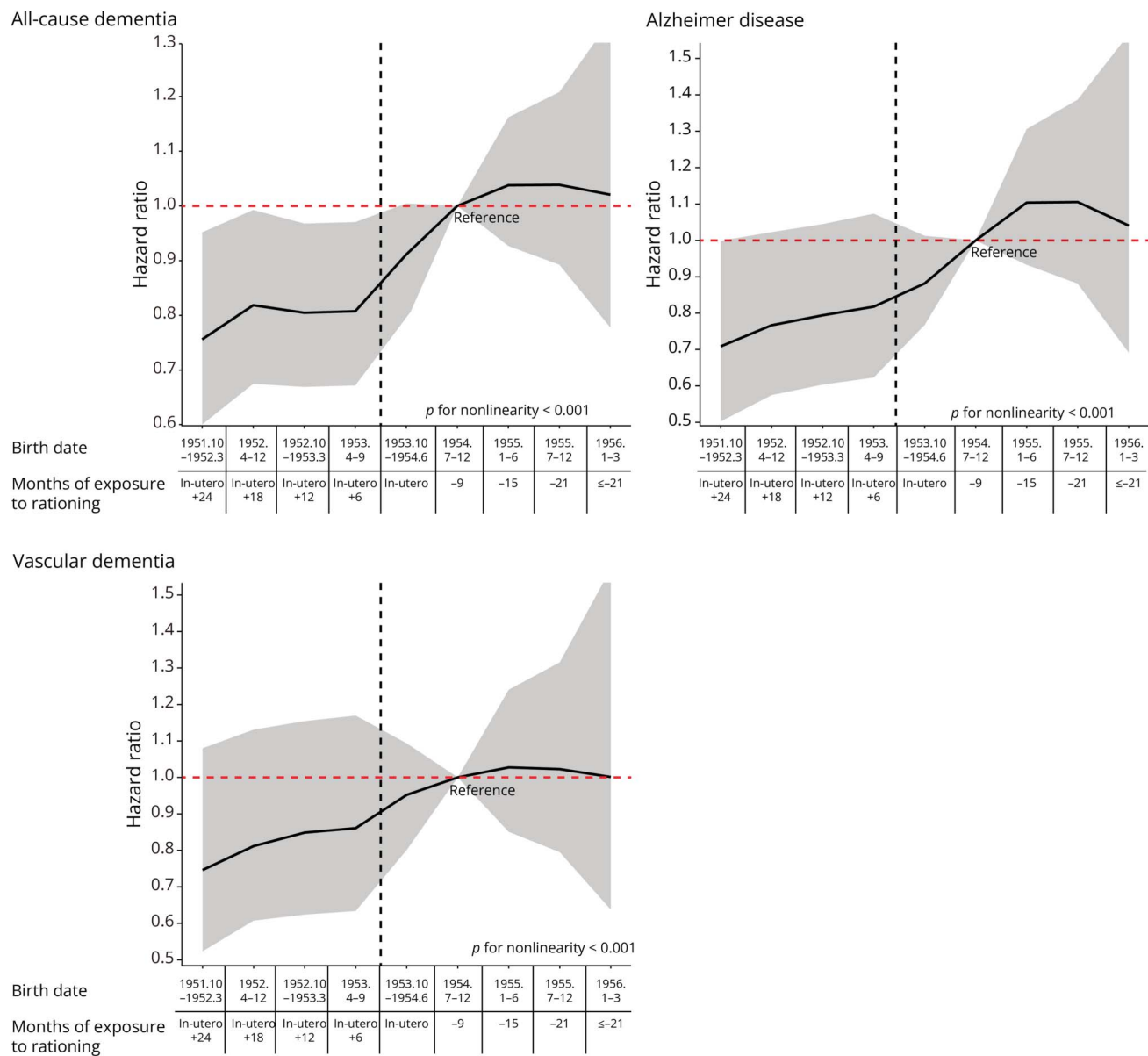
Abbreviations: BMI = body mass index; IQR = interquartile range; MET = metabolic equivalent of task. Data are presented as No (%), mean (SD), or median (IQR). *p* values were obtained from either a χ^2 test or a Mann-Whitney *U* test comparing difference between rationed and not rationed group.

Table 2 Risks of Dementia Subtypes by Sugar Rationing Exposure Duration

	Not rationed	In utero	In utero + (0, 1] year	In utero + (1, 2] year	<i>p</i> for trend
All-cause dementia					
Total cases/total sample size	504/23,774	251/10,682	388/15,025	456/15,256	
Model 1 HR (95% CI)	Reference	0.87 (0.74–1.03)	0.80 (0.67–0.95)	0.78 (0.63–0.96)	0.014
Model 2 HR (95% CI)	Reference	0.86 (0.73–1.01)	0.78 (0.66–0.93)	0.76 (0.61–0.93)	0.007
Model 3 HR (95% CI)	Reference	0.88 (0.75–1.04)	0.82 (0.68–0.97)	0.79 (0.64–0.98)	0.024
Model 4 HR (95% CI)	Reference	0.87 (0.74–1.03)	0.79 (0.66–0.94)	0.77 (0.63–0.95)	0.012
Alzheimer disease					
Total cases/total sample size	227/23,774	112/10,682	179/15,025	206/15,256	
Model 1 HR (95% CI)	Reference	0.84 (0.66–1.08)	0.79 (0.61–1.03)	0.73 (0.53–0.99)	0.052
Model 2 HR (95% CI)	Reference	0.83 (0.65–1.06)	0.77 (0.60–1.01)	0.71 (0.52–0.96)	0.033
Model 3 HR (95% CI)	Reference	0.86 (0.67–1.10)	0.81 (0.62–1.05)	0.75 (0.54–1.03)	0.085
Model 4 HR (95% CI)	Reference	0.84 (0.65–1.07)	0.77 (0.59–1.00)	0.72 (0.53–0.98)	0.041
Vascular dementia					
Total cases/total sample size	178/23,774	95/10,682	135/15,025	157/15,256	
Model 1 HR (95% CI)	Reference	0.94 (0.72–1.23)	0.80 (0.60–1.08)	0.77 (0.54–1.10)	0.121
Model 2 HR (95% CI)	Reference	0.93 (0.71–1.22)	0.79 (0.59–1.06)	0.75 (0.52–1.06)	0.082
Model 3 HR (95% CI)	Reference	0.95 (0.72–1.25)	0.81 (0.60–1.09)	0.78 (0.54–1.12)	0.136
Model 4 HR (95% CI)	Reference	0.95 (0.72–1.25)	0.80 (0.59–1.07)	0.77 (0.54–1.10)	0.113

Abbreviations: DAG = directed acyclic graph; HR = hazard ratio; NFS = National Food Survey. Model 1 to model 3 were cox proportional hazard model. In model 1, we adjusted for age and sex. In model 2, we included DAG-selected covariates, including age, sex, race, birth location, calendar month of birth, real food prices (adjusted for the consumer price index), parental dementia, whether parents still alive, maternal smoking around birth, whether breastfed as a baby, the social average fat intake around the period of birth (derived from the NFS), and survey year. In model 3, we further adjusted household income, Townsend deprivation index, baseline lifestyle factors (smoking status, alcohol intake, and physical activity), and baseline medical conditions (cardiovascular disease, hypertension, diabetes, digestive disease, kidney disease, and liver disease). Model 4 adjusted for terms in Model 3 and used parametric hazard models based on the Gompertz distribution.

Figure 2 Hazard Ratios for Various Types of Dementia Across Different Levels of Exposure to Rationing



Parametric hazard models based on the Gompertz distribution were applied. Each point represents a hazard ratio that compares the hazard rate of adults exposed to rationing during different periods: in utero, in utero plus 6, 12, 18, or 24 months after birth, or not exposed to rationing, with a reference group consisting of adults born between July and December 1954. The group with no exposure to rationing includes individuals born between January and June 1955 (at least 15 months after rationing), July and December 1955 (at least 21 months after rationing), and January to March 1956. Hazard ratio estimates for adults born between January 1955 and April 1956, who were never exposed to rationing, did not show significant differences from those of the reference group ($p = 0.483, 0.520, \text{ and } 0.589$ for all-cause dementia, Alzheimer disease, and vascular dementia, respectively). The model was adjusted for age, sex, race, birth location, calendar month of birth, real food prices (adjusted for the consumer price index), parental dementia, whether parents still alive, maternal smoking around birth, whether breastfed as a baby, the social average fat intake around the period of birth (derived from the NFS) and survey year. The shaded area represents the 95% CI. The vertical dashed black line marks the end of sugar rationing. NFS = National Food Survey.

sugar rationing effect on dementia, respectively. When these mediators were incorporated jointly, they explained 25.5% of the effect, whereas birth weight contributed only 2.7%.

Contemporaneous Validation and Control Cohorts

Detailed descriptions of the contemporaneous validation and control cohorts are provided in eAppendix 3 in the supplementary material. The results from analyses using

these cohorts are presented in eTables 7–13 and eFigure 9.

Discussion

Using the UK's 1953 sugar rationing policy as a natural experiment, we link quasirandom early-life exposure to sugar restriction with incident all-cause dementia and Alzheimer

Table 3 Neuroimaging and Cognitive Function Outcomes by Sugar Rationing Exposure

Indicator	Not rationed ^a	Rationed ^a	Rationed vs not rationed--coefficient (95% CI) for model 1	p Value	Rationed vs not rationed--coefficient (95% CI) for model 2	p Value	Rationed vs not rationed--coefficient (95% CI) for model 3	p Value
Neuroimaging outcomes								
Total brain volume, cm ³ (n = 7,724)	1,162.9 ± 106.2	1,162.5 ± 108.0	−1.12 (−6.69 to 4.44)	0.692	0.42 (−5.11 to 5.96)	0.880	1.20 (−4.36 to 6.76)	0.673
Total white matter, cm ³ (n = 7,724)	544.4 ± 59.3	543.6 ± 61.4	−0.73 (−3.85 to 2.37)	0.644	0.001 (−3.10 to 3.10)	0.999	0.43 (−2.69 to 3.54)	0.788
Total gray matter, cm ³ (n = 7,724)	618.5 ± 51.5	618.9 ± 51.5	2.11 (−0.69 to 4.91)	0.141	2.92 (0.13 to 5.72)	0.040	3.27 (0.46 to 6.07)	0.022
Total hippocampus, cm ³ (n = 7,719)	7.76 ± 0.82	7.75 ± 0.83	−0.022 (−0.072 to 0.028)	0.385	−0.011 (−0.061 to 0.039)	0.678	−0.011 (−0.062 to 0.039)	0.662
Left hippocampus, cm ³ (n = 7,719)	3.83 ± 0.45	3.83 ± 0.46	−0.005 (−0.033 to 0.023)	0.739	0.002 (−0.026 to 0.030)	0.897	0.001 (−0.027 to 0.029)	0.942
Right hippocampus, cm ³ (n = 7,719)	3.94 ± 0.46	3.92 ± 0.47	−0.017 (−0.046 to 0.011)	0.229	−0.012 (−0.041 to 0.016)	0.391	−0.012 (−0.041 to 0.016)	0.401
White matter hyperintensities, cm ³ (n = 7,513)	3.84 ± 4.51	3.72 ± 5.61	−0.56 (−0.90 to −0.23)	<0.001	−0.61 (−0.93 to −0.27)	<0.001	−0.64 (−0.97 to −0.31)	<0.001
Cognitive function								
Complex processing speed (n = 9,917)	19.25 ± 4.72	19.31 ± 4.81	0.15 (−0.12 to 0.42)	0.287	0.26 (−0.006 to 0.53)	0.060	0.27 (0.006 to 0.54)	0.040
Verbal and numerical reasoning (n = 20,612)	6.40 ± 2.14	6.42 ± 2.13	0.04 (−0.06 to 0.14)	0.400	0.08 (−0.01 to 0.17)	0.081	0.10 (0.01 to 0.19)	0.027
Nonverbal reasoning (n = 9,931)	8.10 ± 2.06	8.11 ± 2.08	0.04 (−0.07 to 0.16)	0.454	0.12 (0.012 to 0.24)	0.029	0.13 (0.022 to 0.25)	0.018
Working memory (n = 6,404)	6.81 ± 1.31	6.79 ± 1.25	−0.04 (−0.16 to 0.07)	0.459	−0.04 (−0.15 to 0.07)	0.469	−0.03 (−0.15 to 0.079)	0.561
Verbal declarative memory (n = 9,979)	6.94 ± 2.51	6.87 ± 2.57	−0.08 (−0.22 to 0.058)	0.250	0.004 (−0.13 to 0.14)	0.957	−0.01 (−0.15 to 0.13)	0.881
Trail making test A (n = 9,912)	221.7 ± 73.4	221.5 ± 77.9	−3.75 (−8.03 to 0.51)	0.085	−4.62 (−8.89 to −0.36)	0.033	−4.53 (−8.82 to −0.25)	0.038
Trail making test B (n = 9,723)	547.7 ± 217.8	553.3 ± 253.6	−2.88 (−16.5 to 10.8)	0.678	−9.29 (−22.68 to 4.09)	0.173	−8.04 (−21.4 to 5.39)	0.241

Abbreviations: DAG = directed acyclic graph; NFS = National Food Survey.
^aValues are displayed as mean ± SD. We estimated the coefficient using linear model. In model 1, we adjusted for age and sex. In model 2, we included DAG-selected covariates, including age, sex, race, birth location, calendar month of birth, real food prices (adjusted for the consumer price index), parental dementia, whether parents still alive, maternal smoking around birth, whether breastfed as a baby, the social average fat intake around the period of birth (derived from the NFS) and survey year. In model 3, we further adjusted household income, Townsend deprivation index, baseline lifestyle factors (smoking status, alcohol intake, and physical activity), and baseline medical conditions (cardiovascular disease, hypertension, diabetes, digestive disease, kidney disease, and liver disease).

disease, validated in 2 independent UK/US cohorts. Exposure to rationing in utero plus 1–2 years reduced all-cause dementia and Alzheimer disease hazards by ~23–28%, delayed disease onset by up to 2.8 years. The protective effects strengthened with longer exposure. Diabetes and hypertension jointly mediated 25.5% of this association, whereas birth weight contributed only 2.7%. Furthermore, participants in the sugar rationing group had higher gray matter volume and fewer white matter hyperintensities. Rationed group also showed improved performance in processing speed, reasoning, and memory. Placebo tests and external validation/

control cohorts further confirmed the specificity and robustness of the results.

The 2024 report of the Lancet standing Commission presents 14 potentially modifiable risk factors (beginning with early-life education) and estimates that nearly half of dementia cases may be attributable to these factors.²⁶ To the best of our knowledge, our study is the first to observe a protective effect of sugar rationing during the first 1,000 days of life on the long-term risk of dementia. Previous studies have shown that excessive maternal sugar intake is associated with adverse cardiometabolic

outcomes, including impaired glucose regulation (e.g., gestational diabetes/insulin resistance), dyslipidemia, and higher adiposity-related metabolic risk.²⁷⁻³⁰ Gracner et al.¹⁰ indicated that sugar rationing may provide protective effects against diabetes and hypertension. While high sugar dietary pattern has been shown to negatively affect cognition in older adults, the effect of in-utero and infant sugar restriction on dementia in adulthood has yet to be explored. We extended the “fetal origins of disease” hypothesis to dementia, suggesting the potential of early-life sugar restriction in shaping cognitive aging and dementia prevention strategies.

In this study, participants exposed to sugar rationing during early-life periods showed better neuroimaging outcomes and cognitive function. These findings align with animal study showing exposure to excess sugar during the prenatal periods may lead to adverse changes in the extracellular fluid collected in the offspring’s medial prefrontal cortex and hippocampus.³¹ In human studies, maternal intake of both total and added sugars during pregnancy, particularly in the second trimester, has been linked to alterations in infant brain structure.⁴ In addition, research has shown that maternal fructose intake during the first month postnatally is inversely correlated with cognitive development scores in infants at 24 months.³ Building on previous research, our results suggested the long-lasting beneficial cognitive effects of early-life sugar restriction, reinforcing the protective effects of sugar rationing against dementia observed in our findings.

We observed that the protective effects of early-life sugar rationing against dementia were more pronounced in women. This may be related to gender differences in neural activity. Animal models support our observation: in female mice, the proportion of active neurons in brain regions involved in cognitive and emotional regulation circuits (e.g., the bed nucleus of the stria terminalis) is more than twice that seen in males.³² Consequently, female brains may exhibit a stronger response to early nutritional influences, leading to more pronounced effects on cognitive function. By contrast, no significant differences were observed across *APOE4* genotypes. This may reflect the independence of sugar metabolism interventions from A β /Tau pathology and aligns with previous studies that highlighted the broad benefits of supplement intake across different *APOE* genotypes.³³ These findings suggest that early-life sugar rationing could be an effective, non-genotype-specific strategy for reducing dementia risk, especially in female populations.

Our mediation analysis suggested that although type 2 diabetes and hypertension jointly explain a 25.5% of the sugar rationing-dementia association, the majority of the association may go beyond the pathways of diabetes and hypertension. Although the exact mechanisms remain unclear, several potential pathways have been suggested by previous evidence. First, excessive maternal sugar consumption can lead to metabolic disturbances,³⁴ such as elevated glucose levels and insulin resistance, which may impair placental function and

nutrient delivery to the developing fetus,³⁵ ultimately adversely affecting brain development. Second, excessive sugar consumption during pregnancy has been linked to increased levels of systemic inflammation, which can cross the placenta and affect fetal brain development.³⁶ Inflammatory cytokines can interfere with neurogenesis and synaptic plasticity, processes essential for cognitive function. Sugar restriction during pregnancy may attenuate these adverse changes, slow cognitive decline, and help prevent dementia in later years.

Complementary neurobiological pathways may also be involved. Early-life nutrition influences key neurodevelopmental processes, including synaptic maturation and myelination, which can shape long-term cognitive reserve.³⁷ In addition, insulin resistance has been implicated in dementia-related neurodegeneration through disrupted insulin signaling in the brain,³⁸ and cardiometabolic dysregulation can accelerate cerebral small vessel disease, a major contributor to white matter hyperintensities and vascular cognitive impairment.³⁹ These mechanisms provide biological context for our observation of lower white matter hyperintensity burden and better cognitive performance among individuals exposed to early-life sugar rationing.

From a public health perspective, these findings are most relevant to populations with high added/free sugar exposure during pregnancy and early childhood, particularly in socio-economically disadvantaged settings. Strategies that may yield the largest benefit include guideline-consistent approaches to reduce population sugar availability and intake, such as taxes on sugar-sweetened beverages that incentivize reformulation and reduce purchasing, restrictions on marketing of high-sugar foods to young children, and prenatal and early-life nutrition support. Potential adverse effects of sugar-reduction strategies include regressive financial impacts of price-based policies and unintended substitution toward other energy-dense foods or non-nutritive sweeteners; equity-focused design, such as using revenue to support maternal-child nutrition, may mitigate these risks.

This study has several strengths. First, we used a large and diverse cohort with a wide range of demographic variables and comprehensive health data, allowing us to adjust for numerous potential confounders. Second, our use of the 1953 UK sugar rationing policy as a natural experiment enabled us to explore the long-term effects of early-life sugar exposure in a real-world setting. Third, the study employed rigorous analytical approaches, placebo tests, external validation and control cohorts, and a wide range of neuroimaging and cognitive function outcomes. Finally, the sugar intakes during sugar rationing aligned with today’s guidelines from the World Health Organization (WHO), US Dietary Guidelines, and the American Heart Association (AHA),⁴⁰⁻⁴² ensuring its relevance.

Limitations are shown in eAppendix 4 in the supplementary material.

This study observed that early-life sugar rationing during the first 1,000 days after conception was associated with lower dementia risk in adulthood, with longer exposure associated with a greater degree of protection. Sugar rationing also delayed the onset of various types of dementia and was associated with improved neuroimaging outcomes and cognitive function. These findings extend the “fetal origins of disease” hypothesis to the cognitive domain. Pragmatic randomized trials of feasible, guideline-consistent interventions that reduce added/free sugar intake during pregnancy and early childhood (e.g., structured dietary counseling and family-level support) are warranted, ideally assessing intermediate neurodevelopmental, cardiometabolic, and brain-health end points with longer-term follow-up where possible. The sugar intake during rationing aligned with current WHO, US Dietary Guidelines, and AHA recommendations, providing evidence to inform taxes on sugar-sweetened beverages and infant food policies in the UN 2030 Agenda.

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Author Contributions

J. Zheng: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. G.Y.H. Lip: study concept or design; analysis or interpretation of data. Z. Zhang: study concept or design. S.W.R. Lee: study concept or design.

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